

Research Base for Liquid-Sector Episodic Pivots and Post-Catalyst Continuation in Crypto Perpetual Futures

Perpetuals are the dominant crypto derivatives venue, and the literature points to a narrower set of robust systematic ideas than a generic “all-alt momentum” playbook. In the academic and institutional work that is closest to your use case, time-series continuation survives better than broad cross-sectional rotation once implementability matters; winner-side effects are stronger than loser-side effects; and large-cap crypto momentum is real but crash-prone. That combination argues for leader continuation and post-catalyst base setups in liquid perpetuals, not tiny order-book scalps and not unconditional laggard rotation. ¹

That preference also fits the market structure of perpetuals themselves. Crypto event studies face slower and noisier information incorporation than equities, crypto trades continuously, and prices are fragmented across venues. On derivatives venues, mark price and premium-index mechanics explicitly depend on index construction, funding, and order-book-based measures, which means the shorter the horizon, the more you are leaning on market plumbing rather than on a durable repricing mechanism. Slower episodic continuation setups are less dependent on historical depth than intraday scalps that try to mine tiny microstructure edges. ²

The practical conclusion is straightforward. The best-supported primary families for your research base are **sector ignition leader longs** and **post-catalyst continuation base longs**. **Failed-catalyst shorts** belong in the stack, but as a secondary setup that waits for structural failure. **Sector laggard catch-up** should be optional and tightly constrained, because the liquid-crypto literature is much more comfortable with winners than with losers. ³

Empirical base

Underreaction, overreaction, and what that implies

Crypto markets are generally less informationally efficient than stock markets, and event-study practitioners explicitly warn that crypto prices may not absorb information as quickly as public equities. That matters because it creates room for post-event drift and for setups that wait for a base rather than paying event-day spread and crowding. ⁴

There is direct crypto evidence for both underreaction and overreaction, but the distinction depends on mechanism. A 2024 study in the *Journal of Behavioral and Experimental Finance* finds that abnormal Google search attention predicts higher subsequent crypto returns, higher volatility, and higher trading volume, with only partial reversal afterward; the effect is stronger among larger cryptocurrencies. A 2024 *PLOS One* study likewise finds that investor attention positively influences Bitcoin returns and that reversal dynamics matter across subsamples. A newer crypto adaptation of the frog-in-the-pan idea reports that more

continuous price paths predict stronger future returns, consistent with underreaction to gradual information arrival rather than to one-shot headline spikes. ⁵

That split is important for strategy design. If information arrives gradually through breadth expansion, sector leadership, and improving relative strength, continuation is plausible. If price moves are mostly attention shocks, the path is more fragile: attention predicts near-term strength, but the mechanism can be temporary and prone to reversal. This is exactly why catalyst classification should be based on **what changed mechanically** rather than on whether the headline looked “good” or “bad.” ⁵

Listings, listings with leakage, and why day-zero chasing is weak

The best-established crypto event effect is the exchange-listing effect. Ante’s study of 327 listings across 22 exchanges finds a significant average abnormal return of **5.7% on the listing day** and **9.2% from three days before to three days after**. But that same paper also shows substantial heterogeneity: some exchanges show no significant effect or even negative effects, and there is evidence consistent with informed trading or manipulation before the event. ⁶

A newer 2025 listing study comparing Binance and Coinbase finds positive short-term listing returns on both venues but materially stronger returns on the more regulated Coinbase, and then extends that result across a broader exchange-regulation index. In other words, “listing” is not one homogeneous catalyst. The destination venue, its user base, and its regulatory standing change the expected payoff. ⁷

For systematic trading, that pushes you away from generic event-day chase longs. Spot-listing access expansion can be real, but leakage is common, pre-runs are common, and some effects are venue-specific. That is better suited to either a **post-event continuation base** or a **failed-catalyst short** than to an unconditional “buy official listing timestamp” rule. ⁸

Sector and narrative rotation in crypto is real, but it is not static

Academic network work shows that the crypto cross-section has time-varying community structure and inter-crypto momentum spillovers. Guo, Hårdle, and Tao explicitly document community patterns among cryptocurrencies and an inter-crypto momentum spillover effect. That is a strong foundation for theme ignition logic, especially when paired with breadth and leader participation. ⁹

Institutional crypto research points in the same direction from a more practical angle. Grayscale’s Crypto Sectors framework shows that leadership rotates across distinct segments over time: in one quarter Financials led on rising centralized-exchange volume and stablecoin adoption themes, while in the next quarter privacy-related assets held up best in a more defensive tape. Grayscale also emphasizes that crypto has “frequent changes in dominant themes and shifts in market leadership,” which is exactly the environment where a sector ignition framework can outperform a static single-name breakout screen. ¹⁰

The catch is that sector structure is partly curated and partly emergent. CoinGecko’s category page notes that some assets overlap across multiple categories, and CoinGecko’s own methodology distinguishes between categories and tags. That means the right infrastructure is not a single frozen sector label. It is a **time-aware primary-use-case sector**, plus **secondary tags**, plus a **data-driven co-movement layer** that checks whether the theme is actually trading as a theme right now. ¹¹

Durable continuation versus sell-the-news

The most durable continuation candidates are the catalysts that change economic substance: access, regulation, supply, or utility. There is good reason to separate these from pure attention shocks. Regulatory event-study work on DeFi tokens finds that major policy interventions tend to create **token-specific repricing** rather than broad systemic moves. That supports a token-level event framework instead of a vague “all regulation is risk-off” heuristic. ¹²

Supply-side events are also highly mechanical. Keyrock’s 16,000-event unlock analysis reports that **90% of unlocks create negative price pressure**, that price weakness often starts before the unlock date, and that team unlocks are the worst offenders. On the positive side, staking and emission design matter economically: NBER work on staking finds that the aggregate staking ratio is central for token reward rates and prices, and that token age and exchange-launch stage matter enough to be explicit controls. A separate 2019 CEPR paper finds a significant long-run relationship between crypto prices and adoption/trustworthiness fundamentals. ¹³

Institutional-access events belong in the same “mechanical” bucket when they widen the buyer base or improve regulated access. Early work on U.S. spot Bitcoin ETFs finds that ETF inflows affected Bitcoin price formation, and a 2025 price-discovery study reports that the most actively traded Bitcoin ETFs led spot price discovery for a large share of the sample. But another 2026 study finds that ETF inflows matter mostly in the short term, with weaker longer-horizon dependence. That argues for treating ETF and institutional-access catalysts as **high quality but horizon-dependent**: often durable enough for continuation trades, but not automatically permanent. ¹⁴

By contrast, the lowest-durability bucket is pure attention. If the only thing that changed is mindshare, your default should be **continuation only if a base forms and breadth confirms**, otherwise classify the move as a sell-the-news candidate. ¹⁵

Catalyst taxonomy

The taxonomy should classify events by **mechanism**, not by headline sentiment. The reason is empirical: listing effects vary by venue, regulation tends to reprice token-specific risk rather than the whole asset class, and attention can move prices without changing access, float, or utility. A systematic database should ask “what channel changed?” before it asks “was the headline bullish?” ¹⁶

Mechanism family	What changed economically	Default continuation bias	Typical failure mode
Exchange access expansion	New spot access, roadmap inclusion, deposits open, trading phases, new geography or regulated venue access	High if it materially widens reachable users and liquidity	Nth listing, leaked event, no breadth, heavy pre-run
Leverage access expansion	New perp, margin, borrow, futures, options, pre-launch futures conversion	Medium and asymmetric	More shortability than new demand

Mechanism family	What changed economically	Default continuation bias	Typical failure mode
Legal or regulatory repricing	Approval, dismissal, settlement, classification, enforcement relief, market-structure clarity	High if the event removes a real overhang	Symbolic policy news with no direct legal effect
Supply shock	Burn, buyback, treasury lock, float reduction, lock extension	High if material versus float and verifiable	Small size, discretionary or non-credible implementation
Unlock or vesting change	Cliff unlock, linear emission, postponement, cancellation, insider/team release	Negative for large unlocks; positive if supply is delayed or reduced	Event already front-run
Protocol utility, fee, or revenue change	Fee switch, burn path, staking reward redesign, paid feature launch, usage shock	High if measurable on-chain and persistent	Governance theater without execution
Major integration or distribution access	Wallet, payments, exchange app, chain integration, default distribution, bridge or oracle inclusion	Medium-high if the integration is user-flow relevant	Marketing partnership with no funnel
ETF or institutional access	ETF approval, broker or custodian access, wealth-platform availability, regulated futures	High for assets with real addressable institutional demand	Anticipated event, flow fades quickly
Attention shock with no durable mechanism	Influencer mention, narrative surge, generic partnership, social buzz	Low by default	Fast reversal after the impulse

The point-in-time state machine should also be explicit. Coinbase's official materials provide clean examples: an asset can move from **roadmap** to **transfer only** to **auction** to **limit only** or **full trading**, and Coinbase warns that timeline length varies from hours to months. On the derivatives side, Coinbase International and OKX both describe **pre-launch or pre-market perpetuals** that later convert to standard perpetual futures once the underlying token launches on spot. Binance futures launch notices provide exact UTC launch timestamps, leverage caps, funding frequency, and a warning that a futures listing does not imply a spot listing. Those state transitions are exactly what makes systematic classification possible. ¹⁷

For backtesting, I would assign each event an **ex-ante durability score** before looking at post-event returns. A simple three-level score is enough at first. **High durability** means a direct change in access, float, legal status, or token economics. **Medium durability** means a real change that can cut both ways, especially leverage access. **Low durability** means narrative or attention with no measurable channel. The important point is that the score must be written at event time, stored immutably, and never overwritten using realized price action. ¹⁸

Time-aware sector map and catalyst database

A usable sector map for crypto perpetuals needs to accept that sector membership is both **overlapping** and **time-varying**. Messari's classification system is hierarchical and explicitly built as a sector/sub-sector/tag taxonomy. CoinGecko distinguishes categories from tags and also notes that a token may overlap across multiple categories. DefiLlama provides category-level protocol groupings and definitions across DeFi verticals, while Binance Research's industry map organizes the ecosystem into verticals, subcategories, and primary use cases. Together, those sources support a hybrid design: one forced **primary sector** for backtests, multiple **secondary tags**, and a separate **data-driven community cluster** inferred from rolling co-movement. ¹⁹

Sector map schema

Field	Purpose
<code>asset_id</code>	Stable internal identifier across symbol migrations
<code>ticker</code> , <code>base_asset</code> , <code>quote_asset</code>	Tradable naming layer
<code>taxonomy_source</code>	Messari, CoinGecko, DefiLlama, Binance Industry Map, internal
<code>taxonomy_version</code>	Snapshot or source revision used at that time
<code>primary_sector</code>	Forced single-use-case label for portfolio construction
<code>sub_sector</code>	Narrower economic grouping
<code>secondary_tags</code>	Multiple tags such as AI, DePIN, privacy, DEX, liquid staking
<code>effective_start_utc</code> , <code>effective_end_utc</code>	Point-in-time validity window
<code>sector_confidence</code>	Confidence in assignment at the time
<code>source_timestamp_utc</code>	When the classification was observed
<code>rationale_text</code>	Human-readable explanation
<code>co_movement_cluster_id</code>	Dynamic cluster from rolling return network
<code>leader_flag</code>	Derived field for large-cap/liquidity leaders inside sector
<code>liquidity_rank_in_sector</code>	Dynamic rank on ADV, OI, and spread
<code>is_overlapping_theme</code>	Boolean to mark non-exclusive narrative tags

The crucial design choice is that **membership must be versioned**. If a token was not called "AI" until a 2025 rebranding wave, you may not assign that label to 2024 history. If a protocol appears in a later industry

map, that is not evidence that it belonged to that category earlier. The safe workflow is daily or weekly snapshots of every external taxonomy, then point-in-time joins in research. ²⁰

The extra co-movement layer matters because curated maps lag markets. The Guo-Härdle-Tao network paper is the best academic reason for this: crypto communities are time-varying, and momentum spillovers travel across those communities. In practice, that suggests using curated membership for **ex ante group definition**, but confirming ignition with **contemporaneous breadth and correlation structure**. ⁹

Catalyst database schema

Field	Purpose
<code>event_id</code>	Immutable event key
<code>asset_id</code>	Token identifier
<code>mechanism_family</code>	One of the taxonomy buckets above
<code>mechanism_subtype</code>	E.g., "Coinbase roadmap," "Binance perp launch," "team unlock"
<code>direction</code>	Long, short, mixed, unknown
<code>event_state</code>	Rumor, roadmap, announced, transfer-only, full-trading, futures-live, executed, dismissed, etc.
<code>first_public_ts_utc</code>	Earliest detectable public disclosure
<code>official_confirm_ts_utc</code>	Timestamp of primary-source confirmation
<code>effective_ts_utc</code>	When the mechanism became active
<code>source_confidence</code>	High, medium, low
<code>primary_source_type</code>	Exchange notice, regulator filing, court order, governance execution, protocol docs, unlock provider
<code>primary_source_ref</code>	Document identifier or URL hash
<code>supporting_source_refs</code>	Secondary corroboration
<code>headline_raw</code>	Original text
<code>classification_note</code>	Why the mechanism was assigned
<code>durability_score_ex_ante</code>	High, medium, low written before trade evaluation
<code>estimated_float_impact_pct</code>	Event size versus free float if applicable
<code>estimated_access_impact</code>	Addressable users/capital channel change if applicable
<code>pre_run_features</code>	Price, volume, funding, OI, breadth ahead of event

Field	Purpose
<code>event_low_px</code> , <code>event_high_px</code> , <code>event_vwap</code>	Price anchors for post-event base logic
<code>unknown_fields_mask</code>	Missingness tracking
<code>entered_by</code> , <code>entered_ts_utc</code>	Audit trail

Handling missing catalyst data is a first-order problem, not a footnote. In crypto, identifying the first public disclosure is often harder than in equities because there is no uniform regulatory channel and the same information may leak through social feeds, blogs, chatrooms, and exchange posts. That means the database should never conflate “**no catalyst**” with “**no catalyst observed**.” Those must be separate states.

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A practical confidence ladder works well here. **High confidence** should mean an official exchange notice, exchange rule page, regulator or court document, protocol governance execution, or source-verified unlock schedule. **Medium confidence** should mean a strong secondary report tied to a primary source. **Low confidence** should mean inferred or uncorroborated information. The main backtest should use high-confidence events only, the first robustness layer can add medium-confidence events, and low-confidence rows should stay out of production research unless they are later promoted. 22

Strategy contracts

Sector ignition leader long

This is the strongest pure-price family in the research set. The support comes from three places at once: liquid-crypto momentum is more convincing on the winner side than the loser side; crypto communities exhibit time-varying spillovers; and institutional sector work shows that dominant themes and leaders rotate over time. The right expression is not “buy all names in the theme.” It is “buy the liquid leader once the theme is clearly trading as a theme.” 23

Contract. Define a point-in-time sector with at least three eligible perpetuals after liquidity filtering. Trigger an ignition when sector breadth expands quickly: for example, at least half of names beat BTC over the prior 24 hours, median relative return is positive, sector volume breadth is positive, and at least one sector leader breaks a 20-day or 30-day high. Enter the **leader**, not the average member, on the first liquid breakout after breadth expansion or on the first successful retest of the breakout level. If BTC and ETH are below trend, or if market breadth is collapsing, skip the signal. This explicitly uses sector breadth as confirmation and BTC/ETH as regime context. 24

Why not equal-weight the whole sector? Because the peer group is noisy and categories overlap. Leaders internalize the theme first, and the literature on realistic crypto momentum says the edge is concentrated among winners. Losers rebound unpredictably enough that they can damage naive cross-sectional rotation. For a production stack, the safest first version is one or two leaders per ignition, scaled by liquidity and volatility, with breadth as the gating variable. 25

Post-catalyst continuation base long

This is the cleanest event-driven family in your universe. It matches the market-structure problem directly: crypto events often drift over multi-day windows, and chasing the first impulse often pays maximum spread and maximum emotional premium. A base entry asks whether the repricing **held** once the first shock passed. ²⁶

Contract. Restrict events to the high- or medium-durability buckets: exchange access expansion, legal or regulatory repricing, supply reduction, utility or fee change, major distribution access, or institutional access. Require an initial event repricing over the first hour to first day. Then require three post-event conditions: price stays above the event low, realized volatility contracts, and funding/OI either normalize or rebuild without major giveback. Enter only on a break above the local base high or event VWAP after at least one base window has formed. Exit if price loses the event low or if the base fails quickly without renewed breadth. ²⁷

This contract is especially appropriate for access and legal catalysts because those often change the reachable buyer set or remove a real overhang, but the full impact need not be exhausted on the first bar. It is also appropriate for utility or fee catalysts when on-chain activity can confirm the mechanism later. It is much less appropriate for attention-only events. ²⁸

Failed catalyst short

This should be in the research program, but not as the lead family. The evidence for shorting “bad news” mechanically in crypto is weaker than the evidence for longing real continuation because the short side combines headline risk, squeezes, and the fact that losers can rebound sharply. The better short is a **failed positive catalyst** that loses structure. ²⁹

Contract. Limit this setup to catalyst types with known leakage, crowding, or temporary attention effects: listings, low-durability partnerships, influencer-driven attention shocks, heavily pre-run leverage-access events, and cliffs or unlock calendars that were widely known in advance. Require post-event failure, not merely “bad vibes”: the token must lose event VWAP or the event low, breadth must fail to confirm, and crowding variables such as funding or OI should look stretched into the event. The cleanest entry is a breakdown from a failed base after the first impulse. ³⁰

The reason to keep this as secondary is simple. A listing or attention spike can still continue when the market regime is hot and sector breadth is strong. The short belongs to **failure**, not to the event category by itself. ¹⁰

Sector laggard catch-up

This is the weakest of the four contracts and should stay optional until the others are built. The liquid-crypto literature does not give strong comfort for a generalized laggard catch-up rule. Han, Kang, and Ryu explicitly report that momentum is concentrated among winners and that losers often rebound in ways that inflict losses on standard cross-sectional strategies. That does not kill laggard catch-up, but it means the trade needs much tighter evidence than the leader trade. ³¹

Contract. Only allow laggard catch-up after a confirmed sector ignition, and only if the laggard is still liquid, has no adverse idiosyncratic event, remains inside the same point-in-time sector, and starts to repair relative strength while sector breadth is still broad. In practice, that means waiting for the laggard to reclaim its own trigger level rather than buying weakness inside a hot theme. If that confirmation is absent, skip it. In a first research cycle, I would cap its portfolio weight well below the leader contract or keep it out of the production candidate set entirely. ³²

Event-study and matched-null design

Crypto event studies need to be intraday and multi-horizon. Because crypto trades continuously and information may be incorporated more slowly than in equities, a single daily close-to-close event window is too blunt for this problem. The clean design is to timestamp every event in UTC and evaluate windows such as 1 hour, 4 hours, 24 hours, 3 days, 5 days, and 10 days. ²⁶

The basic event-study question is still standard: did the asset move abnormally around the catalyst? But here you need a more demanding version. The first comparison is **event-day chase versus post-event base entry**. The second is **catalyst continuation versus generic breakout controls**. The third is **real event versus matched pseudo-event**. Those three comparisons tell you whether your edge is coming from the catalyst, from ordinary trend mechanics, or from market regime alone. ³³

A workable design is the following. For each event, estimate raw and abnormal returns using a crypto-specific expected-return model that includes BTC, ETH, and point-in-time sector exposure rather than an equity index benchmark; Brattle's crypto event-study note explicitly warns against naïvely importing stock benchmarks. Then split the sample by catalyst family, durability score, pre-run, sector breadth, BTC/ETH regime, funding/OI state, and listing age. Listing age deserves its own split because token age and exchange-launch stage have explanatory power in the staking literature and because newly launched assets behave differently from seasoned listings. ³⁴

For the **generic breakout control**, create non-event breakout days on the same token where price, volume surge, and relative strength look similar but no classified catalyst is present. Match on pre-run, realized volatility, funding change, OI change, sector breadth, and broader risk regime. If catalyst-conditioned breakouts outperform this matched breakout set, then the catalyst is doing real work beyond generic trend following. If not, your setup is just a renamed breakout system. ³⁵

For the **matched-null design**, I would use three nulls. The first is **same-token placebo timestamps** chosen from non-event days matched on volatility, liquidity, pre-run, and regime. The second is **same-sector peer pseudo-events** at the same clock time when the peer has no catalyst. The third is a **cross-sectional nearest-neighbor null** matched on market cap, ADV, OI, funding, listing age, and sector. The null target is not only post-event return. It should also include breakout failure rate, maximum adverse excursion, time-to-new-high, and holding-period Sharpe. That is what allows you to test whether base entries improve path quality rather than just mean return. ³⁶

The key test matrix should include at least these portfolios:

Portfolio	Entry definition	Main question
Event-day long	Buy immediately after official confirmation or next bar	Is there exploitable drift right away?
Post-event base long	Buy only after hold-above-event-low and base breakout	Does waiting improve expectancy and path quality?
Generic breakout long	Same breakout intensity, no catalyst	Is the catalyst better than normal momentum?
Failed-catalyst short	Short only after structural failure	Does failure, not headline category, drive short alpha?
Sector leader ignition	Leader breakout after breadth expansion	Is breadth-confirmed leader momentum distinct from single-name breakout?
Sector laggard catch-up	Laggard reversal after ignition	Is there any residual catch-up edge after confirmation?

I would make the **post-event base versus event-day entry** test a first-class result, not a side panel. It answers the exact question implied by your mandate: whether a less depth-sensitive continuation entry can outperform the obvious first-bar chase in liquid perps. ³⁷

Bias controls, validation, and data collection plan

Bias controls that matter here

The first bias control is **no future sector membership**. That means snapshotting every external taxonomy as it was known at the time, forcing one primary sector label, preserving overlapping tags, and never backfilling later category changes into earlier history. CoinGecko's own documentation says categories and tags are distinct and that tokens can overlap categories; that is exactly why point-in-time snapshots are necessary. ³⁸

The second is **no future catalyst classification**. Crypto often lacks a single canonical first disclosure, so the database must store first-seen timestamps, official-confirmation timestamps, and effective timestamps separately. Brattle's note is explicit that identifying the first disclosure is harder in crypto than in equities; therefore, the database must preserve this ambiguity rather than resolve it retroactively with hindsight. ²¹

The third is **no current-universe survivorship**. Reconstruct the daily tradable perpetual universe from exchange metadata and notices. Binance's exchange-information endpoint exposes `contractType`, `onboardDate`, and `status`, and Binance's delist-schedule documentation states that delivery or delisting information is reflected through exchange-info metadata after the announcement. That means the research universe can be rebuilt point in time instead of from today's active symbols. ³⁹

The fourth is **unknown-is-not-none**. If a catalyst was missing from your collection pipeline, that row should be marked unknown. This matters most for attention shocks and governance events, where the boundary between absence and unobserved presence is blurry. ²¹

Validation plan

I would validate in four layers. First, run a **chronological walk-forward** with frozen rules and rolling re-estimation of sector maps, leader definitions, and liquidity cutoffs. Second, run **regime splits**: BTC/ETH above or below medium-term trend, broad risk-on versus risk-off, high versus low market breadth, positive versus negative sector breadth, high versus low crowding in funding/OI, and heavy versus light pre-run. Third, run **mechanism splits** across catalyst families, because legal repricing, exchange access, unlocks, and attention shocks should not be pooled as one event class. Fourth, run **exchange splits**, especially for listing- and leverage-related events, because the literature shows venue-specific heterogeneity. ⁴⁰

Path-dependent diagnostics should be central because large-cap crypto momentum is crash-prone. Record maximum adverse excursion, time-under-water, gap risk around announcements, adverse funding carry, and crowding exits. Volatility scaling and exposure caps deserve their own ablations, because the liquid-crypto momentum literature finds that volatility management materially improves crash behavior. ⁴¹

Data collection plan

The core market-data stack should include **trade OHLCV, mark price, index price, funding history, open interest, lifecycle metadata, and delist metadata**. Official Binance documentation is enough to justify the schema: mark price and price index are separate objects, current open interest is available by symbol, funding history is queryable by time range, and exchange information includes onboard date and status. Bybit and Coinbase publish corresponding documentation for funding and mark-price mechanics. ⁴²

There is one operational catch: official exchange endpoints are not always deep enough historically. Binance's open-interest statistics endpoint exposes only the latest one-month history at common bar granularities, so if you want long OI history you should either archive continuously from now on or license a vendor with historical coverage. That limitation should be written into the data lineage from day one. ⁴³

The classification and fundamentals stack should combine **Messari classifications, CoinGecko categories and tags, DefiLlama protocol categories and revenue/fees datasets, Binance Research industry maps, and institutional sector frameworks** such as Grayscale's. DefiLlama also exposes categories, open-interest pages, unlock calendars, and an API. That mix is good enough to build a time-aware sector map and to cross-check category drift. ⁴⁴

The catalyst stack should prefer **official and primary sources**: exchange notices, roadmaps, help pages, regulator or court documents, governance proposals and execution logs, protocol documentation, and source-verified unlock providers. Coinbase roadmap pages, Coinbase listing-phase documentation, Binance futures launch notices, Coinbase pre-launch rules, and OKX pre-market-futures notices are all backtest-friendly because they publish state transitions and timestamps. Unlock calendars can then be layered in from specialized providers and checked against protocol announcements where possible. ⁴⁵

A compact source plan looks like this:

Data block	Primary sources	Notes
OHLCV, trades	Exchange APIs and historical trade feeds	Prefer venue-native data for tradable perps

Data block	Primary sources	Notes
Mark and index	Exchange docs and APIs	Keep both; do not backtest only on last trade
Funding	Official funding-history endpoints	Archive continually
Open interest	Official OI endpoints plus vendor archive	Long history may require vendor or continuous capture
Universe metadata	Exchange info, onboard dates, status, delist schedules	Needed for survivorship-free historical universes
Sector map	Messari, CoinGecko, DefiLlama, Binance Research, Grayscale	Snapshot and version everything
Catalyst database	Official notices, regulatory documents, governance, unlock providers	Store first-seen, official-confirm, and effective timestamps separately
Execution stress	Rule-based slippage and latency grid	Use decision-bar plus next-bar execution assumptions

For execution stress, I would not try to forecast full historical order-book depth if your aim is to move away from microstructure dependence. Instead, stress test with conservative rules: trade on next bar, haircut fills around announcements, cap participation, use mark/index for validation, and widen slippage around high-premium or high-funding conditions. Exchange docs make clear that mark and funding mechanics already encode market-plumbing strain; that is a better robustness target than a fake precision model of historical depth. ⁴⁶

Backtesting-agent prompt

The prompt below is designed to force point-in-time sector and catalyst handling, survivorship-free universe construction, and direct tests of event-day versus post-base entry. It reflects the exchange metadata, category/tag overlap, and disclosure-timestamp issues documented in the sources above. ⁴⁷

You are building a research and backtesting pipeline for crypto perpetual futures with an emphasis on liquid-sector episodic pivots and post-catalyst continuation.

Objective

Build a point-in-time research platform and test suite for these strategy families:

- 1) sector ignition leader long
- 2) post-catalyst continuation base long
- 3) failed catalyst short
- 4) sector laggard catch-up only if evidence supports it

Non-negotiable bias controls

- Never use future sector membership.

- Never use future catalyst classification.
- Never use current-universe survivorship.
- Distinguish “no catalyst observed” from “no catalyst”.
- Store first_public_ts, official_confirm_ts, and effective_ts separately.
- All sector memberships and tags must have effective_start and effective_end timestamps.
- Reconstruct the tradable perp universe point in time from onboard/delist/status metadata.

Research assumptions

- Focus on liquid perpetuals rather than illiquid alts.
- Use slower event/breadth continuation logic rather than microstructure scalp logic.
- Event classes must be classified by mechanism, not by headline sentiment.
- Catalyst families:
 - a) exchange access expansion
 - b) leverage access expansion
 - c) legal/regulatory repricing
 - d) supply shock
 - e) unlock/vesting change
 - f) protocol utility/fee/revenue change
 - g) major integration/distribution access
 - h) ETF/institutional access
 - i) attention shock with no durable mechanism

Data model requirements

A. Instrument master

Fields:

- asset_id
- symbol history
- venue
- contract_type
- onboard_date_utc
- delist_date_utc
- status history
- quote asset
- base asset
- listing age in days
- underlying metadata

B. Market data

Fields:

- bar_ts_utc
- trade OHLCV
- mark OHLCV
- index OHLCV
- funding rate

- open interest
- basis / premium if available
- taker imbalance if available
- execution stress flags

C. Sector map

Create a point-in-time hybrid taxonomy with:

- primary_sector (forced single label)
- sub_sector
- secondary_tags (many-to-many)
- taxonomy_source
- taxonomy_version
- effective_start_utc
- effective_end_utc
- sector_confidence
- rationale_text
- co_movement_cluster_id
- leader_flag
- liquidity_rank_in_sector

D. Catalyst database

Fields:

- event_id
- asset_id
- mechanism_family
- mechanism_subtype
- direction
- event_state
- first_public_ts_utc
- official_confirm_ts_utc
- effective_ts_utc
- source_confidence
- primary_source_type
- primary_source_ref
- supporting_source_refs
- headline_raw
- classification_note
- durability_score_ex_ante
- estimated_float_impact_pct
- estimated_access_impact
- unknown_fields_mask
- entered_ts_utc

Build point-in-time sector membership

- Snapshot every external taxonomy over time.
- Force one primary sector per asset per timestamp.
- Preserve overlapping tags.
- Add a dynamic co-movement cluster using rolling return correlations / lead-lag

network to validate that the theme is trading as a theme right now.

- Never label an asset with a narrative before that narrative classification existed in the raw source history.

Catalyst labeling rules

- Prefer primary sources.
- Confidence ladder:
 - High = official exchange notice, regulator/court filing, governance execution, source-verified unlock provider
 - Medium = reputable secondary source tied to primary source
 - Low = inferred / uncorroborated
- Main backtests use High only.
- Robustness can add Medium.
- Low-confidence events excluded from main results.

Regime context features

Create these context variables per bar:

- BTC medium-term trend
- ETH medium-term trend
- market breadth
- sector breadth
- funding state
- OI state
- liquidity state
- pre-run magnitude
- listing age bucket
- broad risk-on / risk-off indicator

Liquidity and execution filters

- Restrict to liquid perpetuals only.
- Filter by ADV, median spread proxy, OI, and minimum listing age.
- Apply next-bar execution.
- Apply conservative slippage schedule by volatility and event intensity.
- Archive raw assumptions for reproducibility.

Operational strategy contracts

1) Sector ignition leader long

Trigger only when:

- point-in-time sector has breadth expansion
- leader participation is present
- leader is liquid and breaks a meaningful trigger level
- BTC/ETH regime and broad market regime are not hostile

Entry:

- breakout or first successful retest

Exit:

- breakout failure, breadth collapse, time stop, crowding extreme

Research outputs:

- hit rate by sector
- dependence on breadth quality
- leader concentration and crowding diagnostics

2) Post-catalyst continuation base long

Eligible catalysts:

- high or medium durability only

Base logic:

- initial repricing after catalyst
- price holds above event low
- realizes a volatility contraction / base
- funding/OI normalize or rebuild constructively

Entry:

- breakout of base high or reclaim of event VWAP after base formation

Exit:

- loss of event low / failed base / time stop

Compare directly against event-day entry.

3) Failed catalyst short

Only secondary.

Eligible mainly for:

- weak-mechanism events
- listing/access events with heavy pre-run
- attention shocks
- unlock-related setups

Trigger:

- loses event VWAP or event low
- breadth fails
- crowding was stretched into event

Entry:

- failed base breakdown

Exit:

- reclaim of failed level / squeeze risk / time stop

4) Sector laggard catch-up

Treat as optional.

Require:

- prior confirmed sector ignition
- laggard remains liquid
- no adverse idiosyncratic catalyst
- laggard repairs relative strength
- sector breadth still healthy

Weight it smaller than leader trades unless evidence becomes strong.

Event-study design

Run intraday and multi-day windows:

- 1h, 4h, 24h, 3d, 5d, 10d

Measure:

- raw returns
- abnormal returns
- max adverse excursion
- max favorable excursion
- time to new high / low
- breakout failure probability

Use crypto-specific expected-return controls:

- BTC
- ETH
- sector
- matched controls

Do not use plain equity-index benchmarks.

Matched-null design

Build three nulls for each event:

- 1) same-token pseudo-events on non-event days matched on vol/liquidity/pre-run/regime
- 2) same-sector peers at same timestamp with no catalyst
- 3) nearest-neighbor tokens matched on sector, size, liquidity, OI, funding, listing age, and regime

Also compare against generic breakout controls on the same token with no classified catalyst.

Core research questions

- Does catalyst continuation outperform generic breakouts?
- Does post-event base entry outperform event-day entry?
- Which catalyst families are most durable?
- Which catalyst families are mostly sell-the-news?
- Does breadth-confirmed sector ignition outperform single-name breakouts?
- Is laggard catch-up robust enough to keep?

Validation

- Walk-forward only
- No random shuffling across time
- Regime splits
- Exchange splits
- Leave-one-sector-out
- Leave-one-catalyst-family-out
- 2025+ out-of-sample if possible
- Report path-quality metrics, not just mean returns

Required deliverables

- 1) point-in-time sector map tables
- 2) catalyst database tables
- 3) reproducible universe-construction pipeline
- 4) event-study notebook / scripts
- 5) matched-null notebook / scripts
- 6) strategy-contract backtests

- 7) ablation tests
- 8) data-quality report with missingness analysis
- 9) research memo summarizing whether each contract survives costs and bias controls

Output style

- Be explicit about assumptions.
- Flag every place where data are missing or inferred.
- Separate facts from heuristics.
- Never overwrite historical classifications with later knowledge.

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